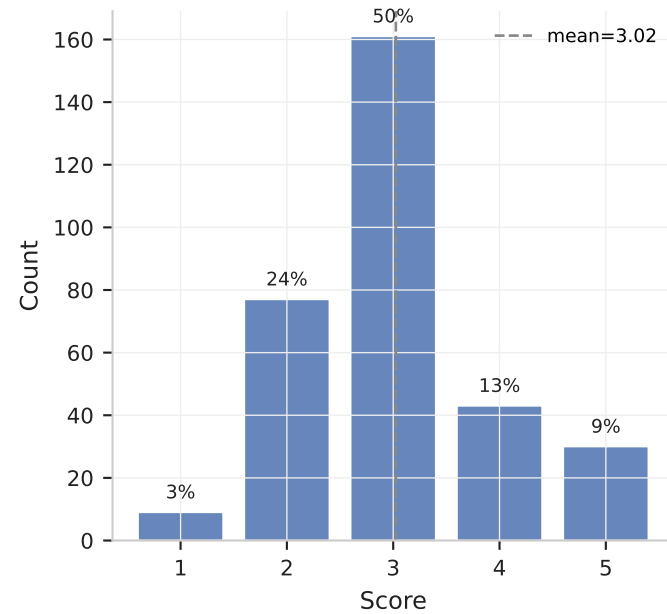
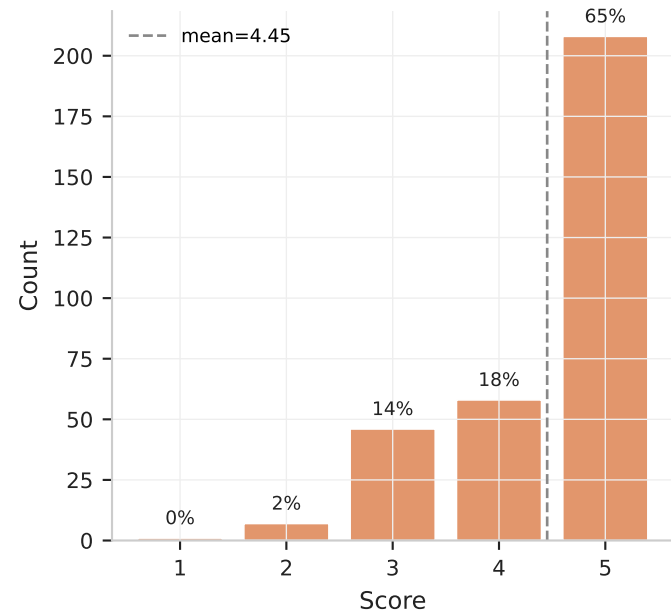


## Score Frequency per Dimension (scores 1-5)

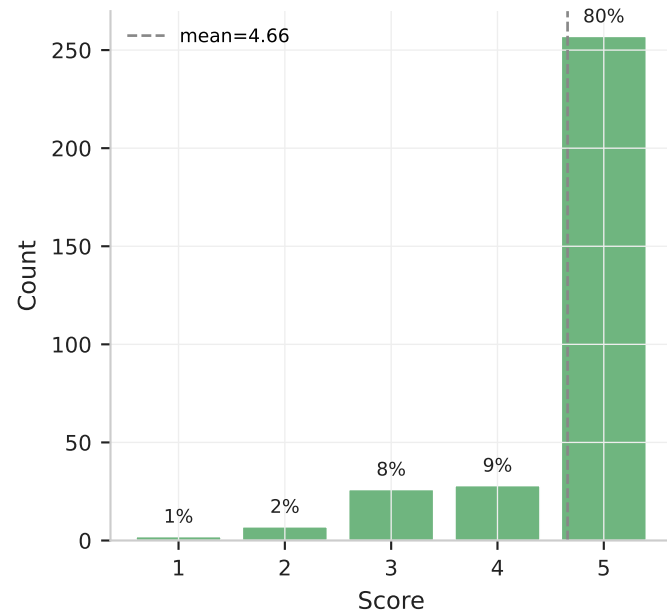
### Socratic Guidance



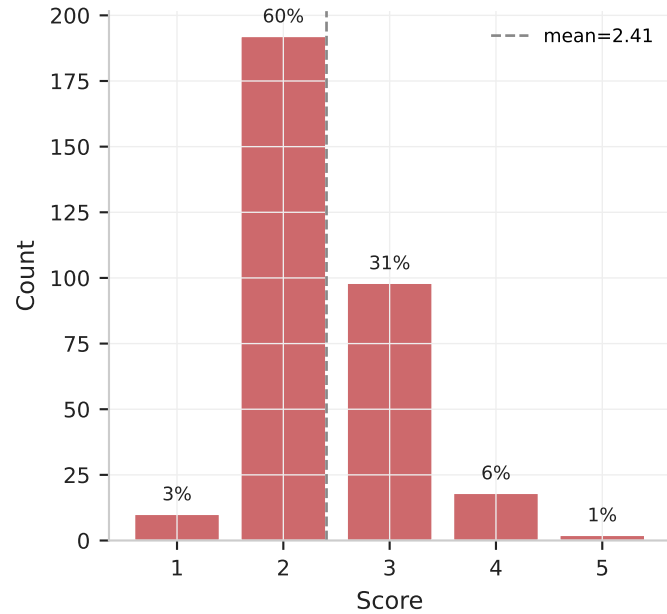
### Mathematical Accuracy



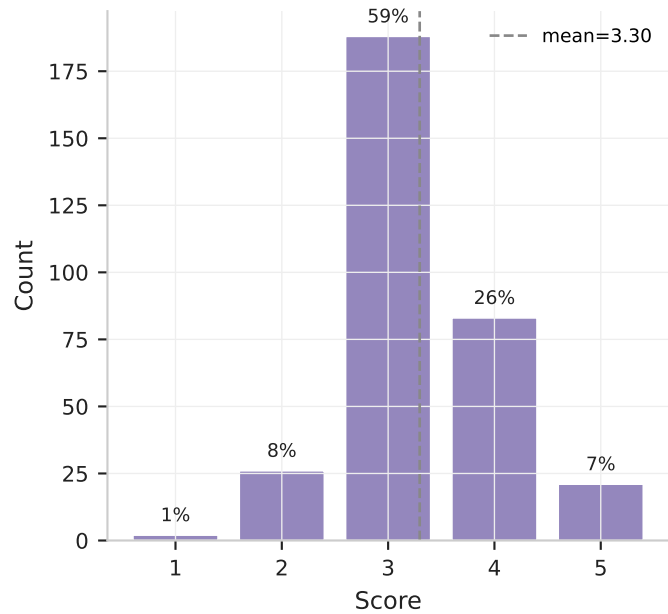
### Relevance



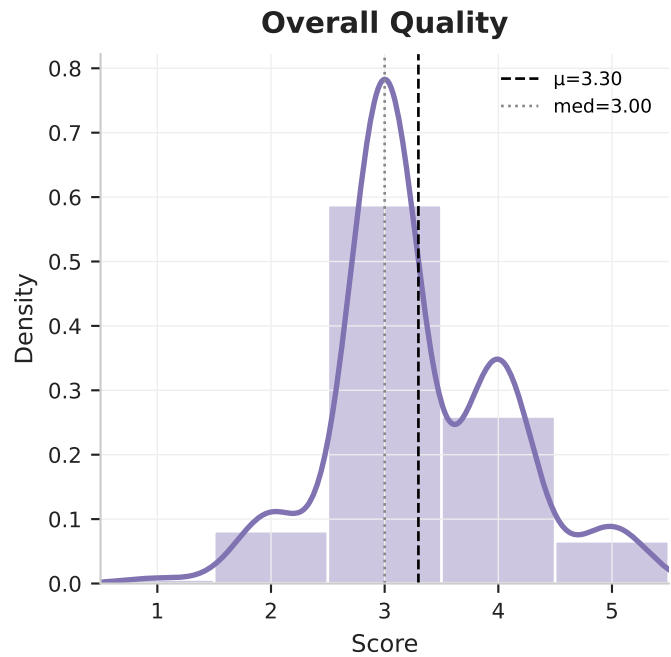
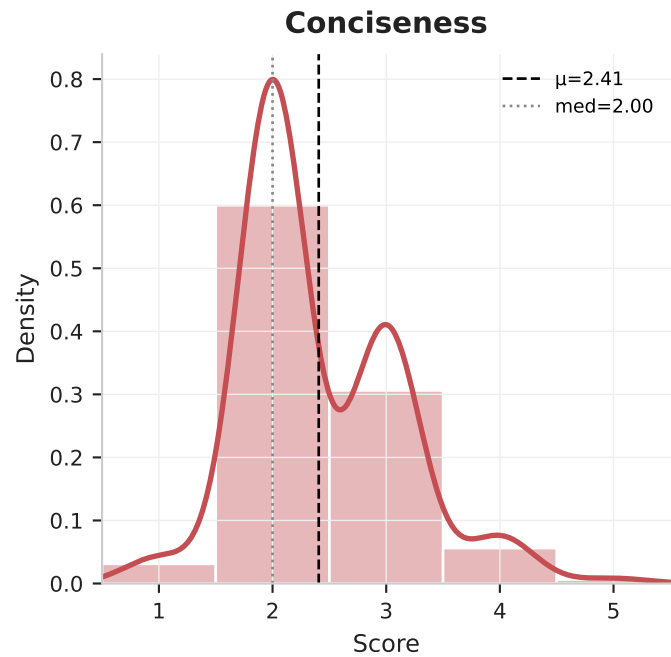
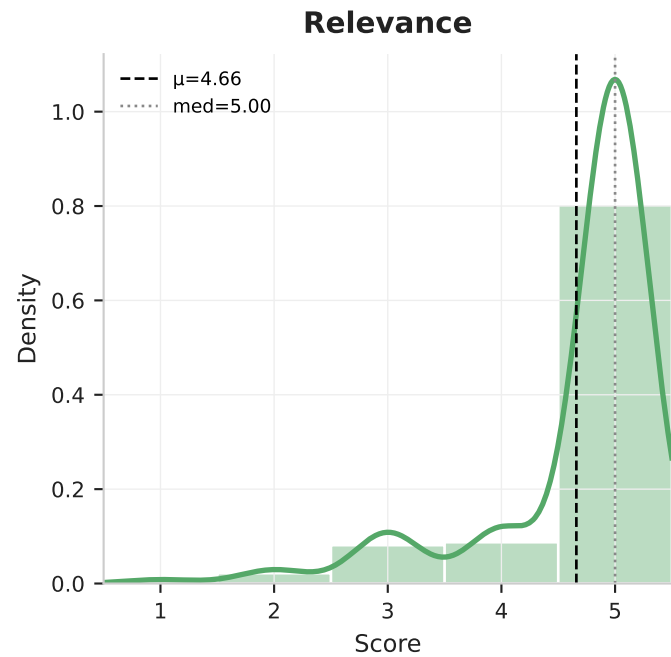
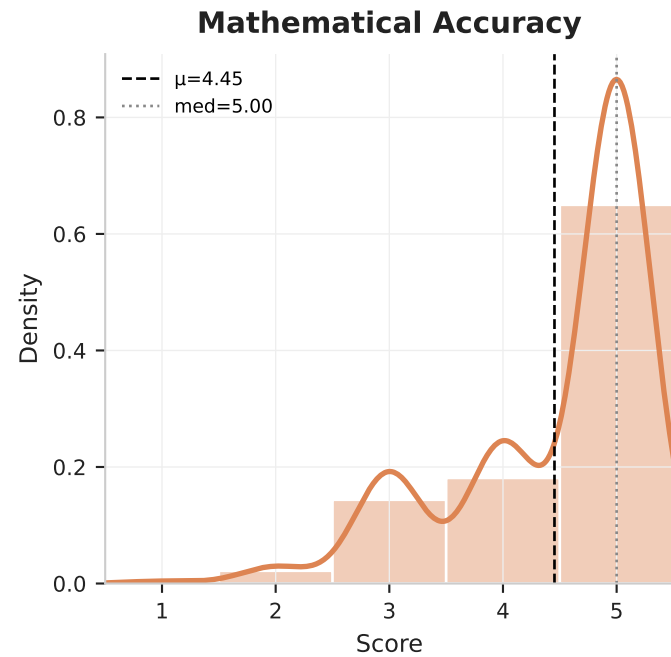
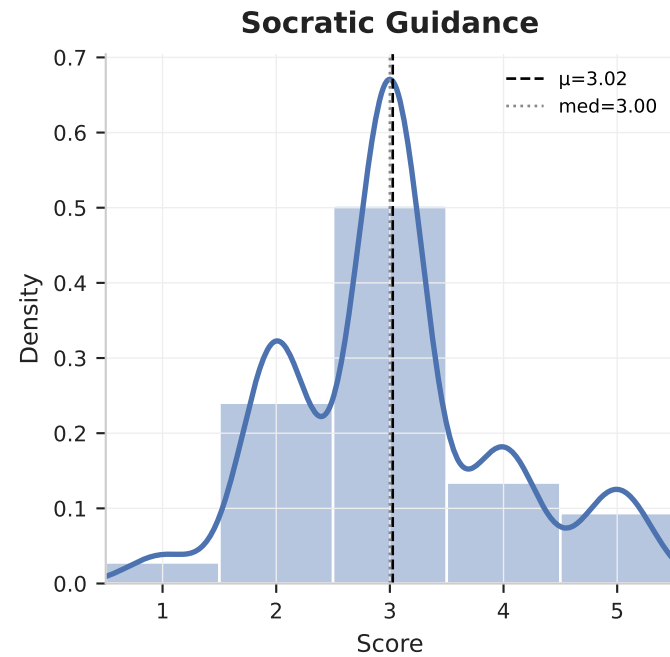
### Conciseness



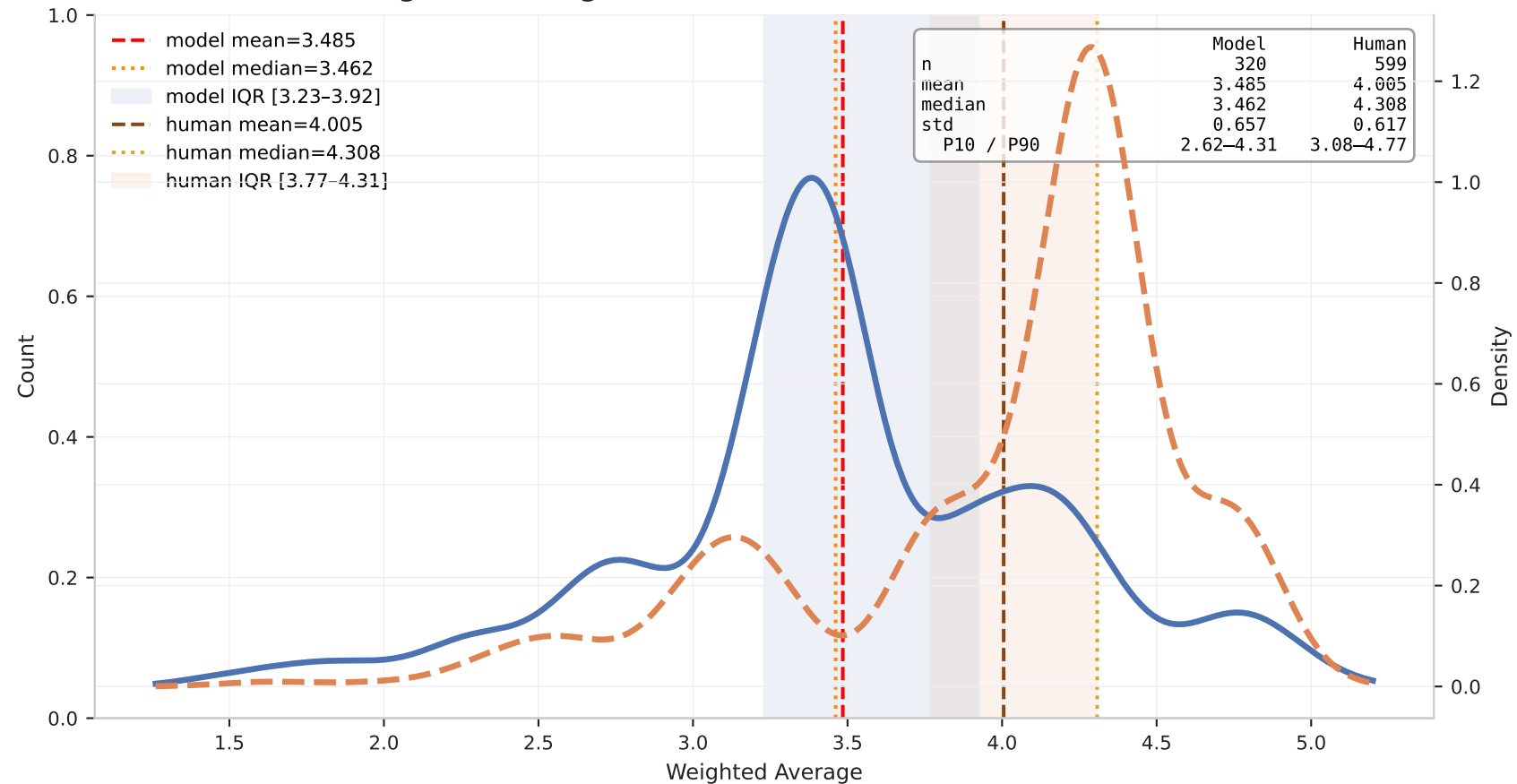
### Overall Quality



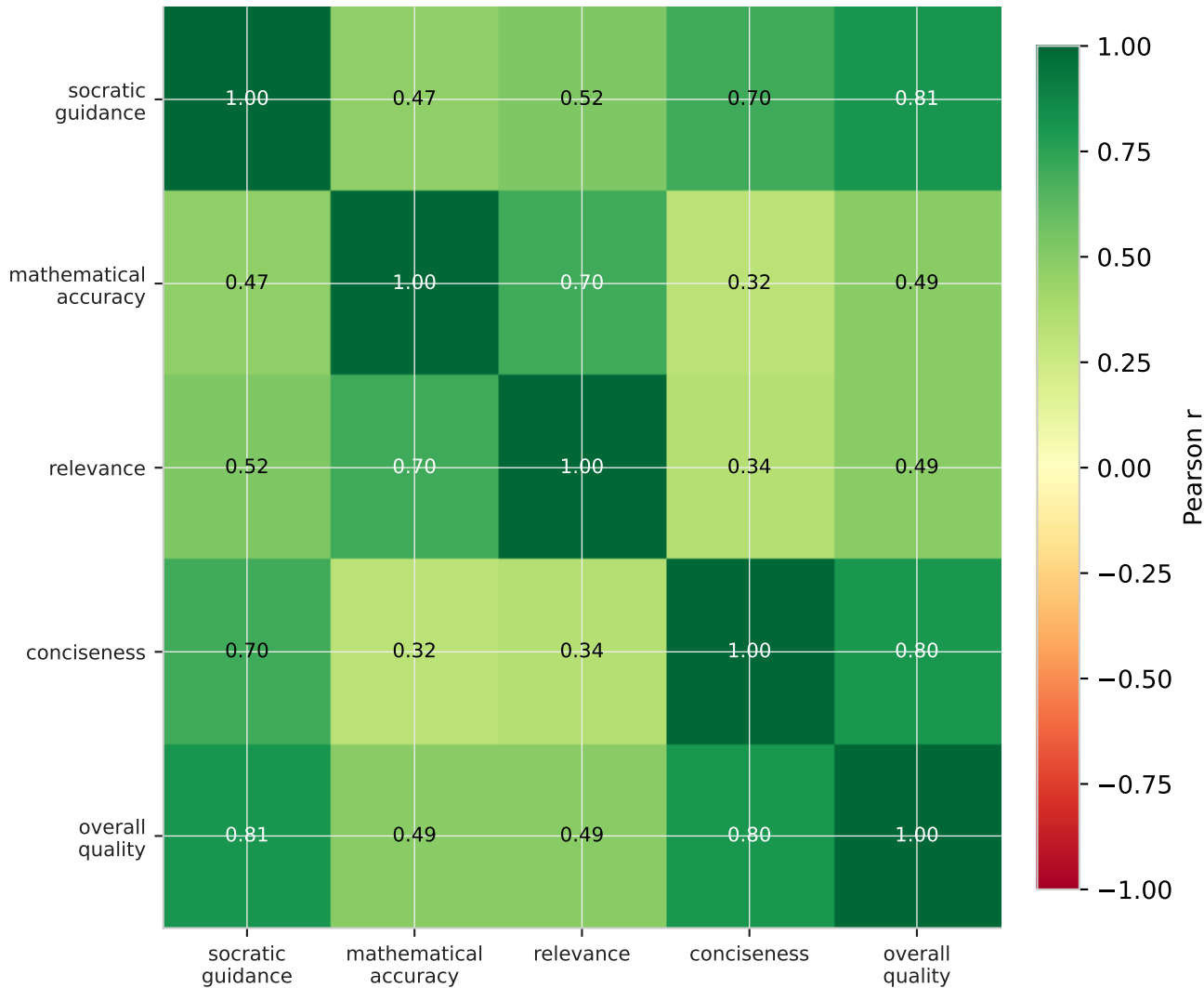
## Score Distribution with KDE (per dimension)



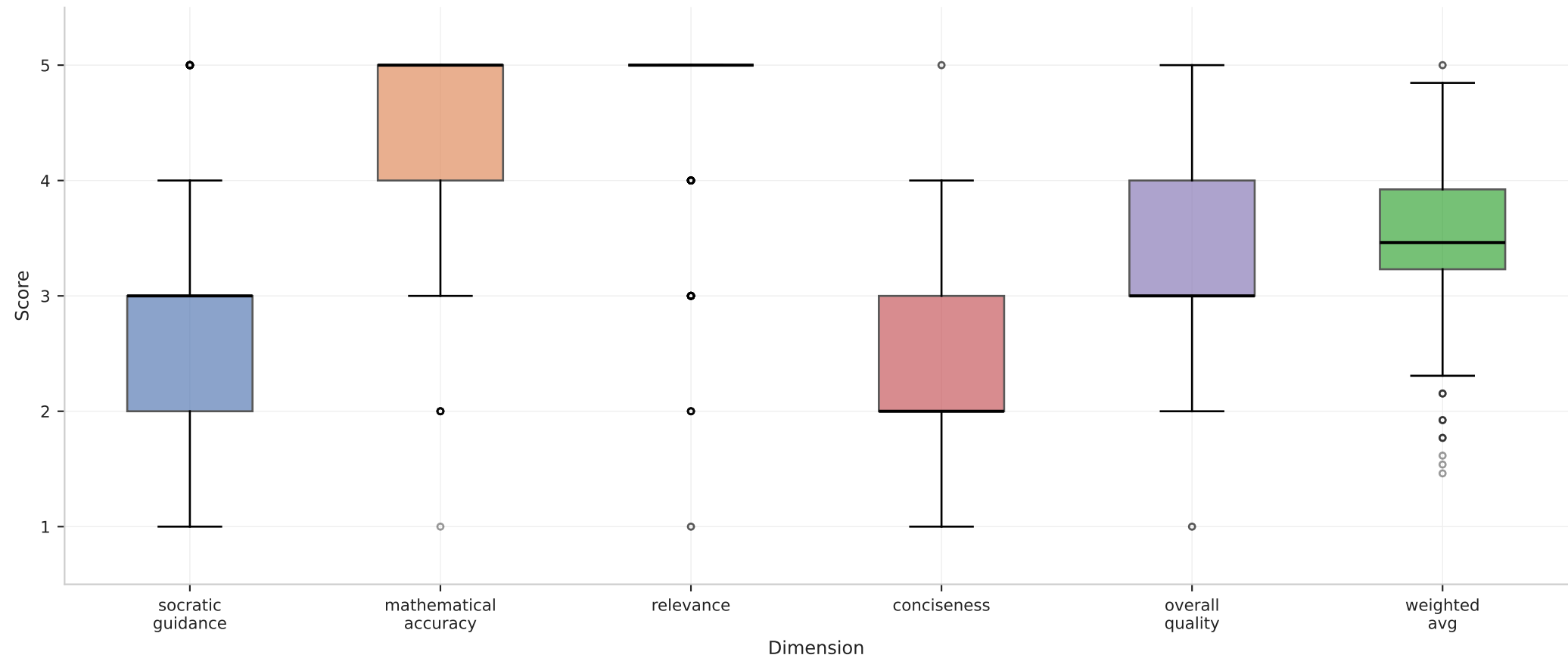
## Weighted Average Score Distribution — Model vs Human



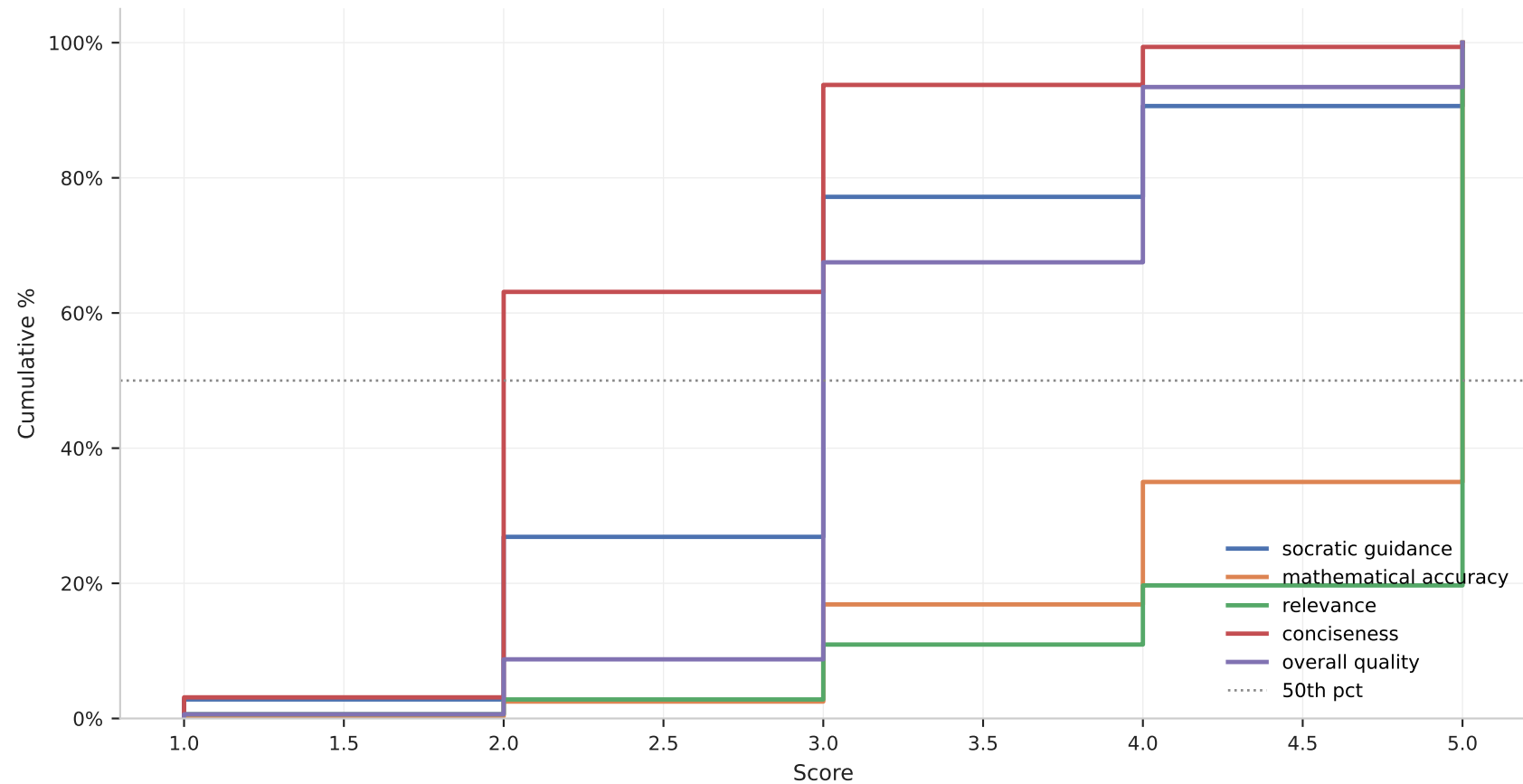
**Cross-Dimension Pearson Correlation Matrix**



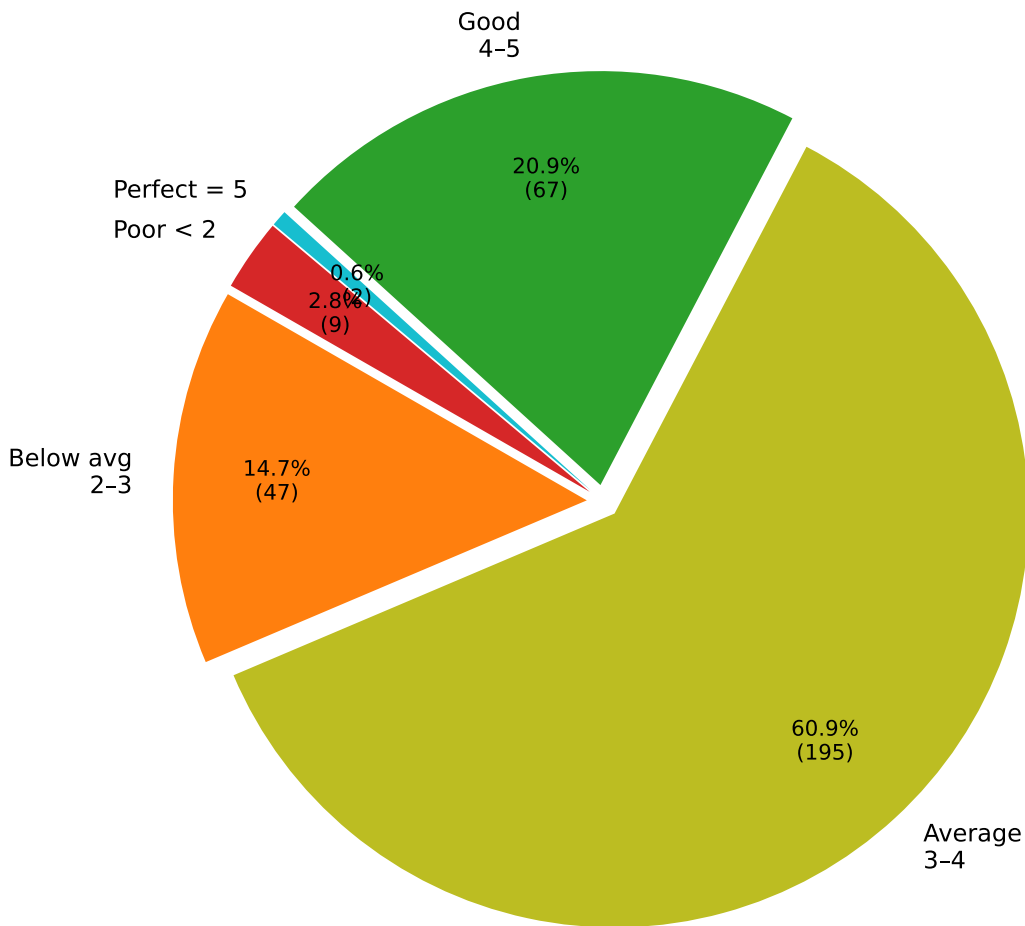
Score Distribution — Box & Whisker (with outliers)



# Empirical Cumulative Distribution Functions



# Grade-Band Distribution (Weighted Average)



Descriptive Statistics Summary

|                       | n   | mean  | median | std   | min  | max  | P25   | P75   |
|-----------------------|-----|-------|--------|-------|------|------|-------|-------|
| socratic guidance     | 320 | 3.025 | 3.000  | 0.928 | 1.00 | 5.00 | 2.000 | 3.000 |
| mathematical accuracy | 320 | 4.453 | 5.000  | 0.839 | 1.00 | 5.00 | 4.000 | 5.000 |
| relevance             | 320 | 4.659 | 5.000  | 0.770 | 1.00 | 5.00 | 5.000 | 5.000 |
| conciseness           | 320 | 2.406 | 2.000  | 0.674 | 1.00 | 5.00 | 2.000 | 3.000 |
| overall quality       | 320 | 3.297 | 3.000  | 0.735 | 1.00 | 5.00 | 3.000 | 4.000 |
| weighted avg          | 320 | 3.485 | 3.462  | 0.657 | 1.46 | 5.00 | 3.231 | 3.923 |